Visual–Inertial Odometry (VIO) Notes

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1 Objective function

We estimate the variables in the active optimization set by minimizing a weighted least-squares cost:

$$J(\mathbf{X}) = \|\mathbf{r}_{\text{prior}}\|^2_{\Sigma^{-1}_{\text{prior}}} + \sum_{(i,j)\in\mathcal{I}} \left\|\mathbf{r}^{\text{imu}}_{ij}\right\|^2_{\Sigma^{-1}_{ij}} + \sum_{(i,\ell)\in\mathcal{C}} \|\mathbf{r}^{\text{cam}}_{i\ell}\|^2_{\Sigma^{-1}_{i\ell}}\,, \tag{1}$$

where $\|\mathbf{r}\|^2_{\Sigma^{-1}} = \mathbf{r}^\top \Sigma^{-1}\mathbf{r}$. The estimate is

$$\hat{\mathbf{X}} = \arg\min_{\mathbf{X}} J(\mathbf{X}). \tag{2}$$

where the optimization variables are collected in

$$\mathbf{X} = \left\{\mathbf{x}_i,\, \mathbf{P}^W_\ell\right\}, \tag{3}$$

and the index sets are:

- $\mathcal{I}$: set of consecutive keyframe pairs (i, j) with an active IMU factor,
- $\mathcal{C}$: set of (keyframe, landmark) pairs (i, ℓ) with a valid reprojection observation,
- $\mathbf{x}_i = (\mathbf{R}_i, \mathbf{p}_i, \mathbf{v}_i, \mathbf{b}_{g,i}, \mathbf{b}_{a,i})$: state at keyframe i,
- $\mathbf{P}^W_\ell \in \mathbb{R}^3$: position of landmark ℓ in the world frame,
- $\mathbf{r}_{\text{prior}}$: residual encoding information from marginalized states,
- $\mathbf{r}^{\text{imu}}_{ij} \in \mathbb{R}^{15}$: IMU residual between keyframes i and j,
- $\mathbf{r}^{\text{cam}}_{i\ell} \in \mathbb{R}^2$: camera reprojection residual for landmark ℓ at keyframe i,
- $\Sigma_{\text{prior}}, \Sigma_{ij}, \Sigma_{i\ell}$: covariance matrices of the corresponding noise terms.

The three terms correspond to: prior information, IMU constraints, and camera reprojection constraints. For simplicity, we omit the prior term in the following.
Intuitively, the prior encodes information from states that have been removed from the active window: when the oldest state is marginalized out, its contribution is compressed into a prior factor that is carried forward, preserving the information without keeping the full state history. Marginalization is thus the process of removing the oldest states while mathematically preserving their information as a prior.

2 State and conventions

At keyframe (or image time) t_i, the state is

$$\mathbf{x}_i = (\mathbf{R}_i, \mathbf{p}_i, \mathbf{v}_i, \mathbf{b}_{g,i}, \mathbf{b}_{a,i}), \tag{4}$$

with $\mathbf{R}_i \in \mathrm{SO}(3)$, $\mathbf{p}_i, \mathbf{v}_i \in \mathbb{R}^3$, and gyro/accelerometer biases $\mathbf{b}_{g,i}, \mathbf{b}_{a,i} \in \mathbb{R}^3$.
We use the convention that $\mathbf{R}_i$ maps vectors from the body frame B_i to the world frame W. Therefore, $\mathbf{R}_i^\top$ maps vectors from world to body coordinates.
With this convention, all frame transformations in the camera and IMU equations stay consistent.

3 Camera model and residual

Let $\mathbf{P}_\ell^W \in \mathbb{R}^3$ be the position of landmark ℓ in the world frame. Let the camera extrinsics be $(\mathbf{R}_{CB}, \mathbf{p}_{BC}^B)$, where:

- $\mathbf{R}_{CB}$ rotates body-frame vectors into camera-frame vectors,
- $\mathbf{p}_{BC}^B$ is the camera position relative to the body origin, expressed in the body frame.

Then the landmark coordinates in the body and camera frames are

$$\mathbf{P}_\ell^{B_i} = \mathbf{R}_i^\top (\mathbf{P}_\ell^W - \mathbf{p}_i), \tag{5}$$
$$\mathbf{P}_\ell^{C_i} = \mathbf{R}_{CB}(\mathbf{P}_\ell^{B_i} - \mathbf{p}_{BC}^B). \tag{6}$$

Using the pinhole model with intrinsic matrix

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \tag{7}$$

and $\mathbf{P}_\ell^{C_i} = (x, y, z)^\top$, the projected pixel is

$$\hat{\mathbf{u}}_{i\ell} = \begin{bmatrix} f_x x/z + c_x \\ f_y y/z + c_y \end{bmatrix}. \tag{8}$$

If $\mathbf{u}_{i\ell}$ is the observed pixel, the camera residual is

$$\mathbf{r}_{i\ell}^{\mathrm{cam}} = \mathbf{u}_{i\ell} - \hat{\mathbf{u}}_{i\ell}. \tag{9}$$

The camera factor penalizes the mismatch between the measured image point and the point predicted by the current state $\mathbf{x}_i$ and landmark estimate $\mathbf{P}_\ell^W$.

4 IMU measurement model

The gyroscope and accelerometer measurements are modeled as

$$\boldsymbol{\omega}_m(t) = \boldsymbol{\omega}(t) + \mathbf{b}_g(t) + \mathbf{n}_g(t), \tag{10}$$
$$\mathbf{a}_m(t) = \mathbf{f}(t) + \mathbf{b}_a(t) + \mathbf{n}_a(t), \tag{11}$$

where

- $\boldsymbol{\omega}(t)$ is the true body angular velocity,

- $\mathbf{f}(t)$ is the true specific force in the body frame,
- $\mathbf{n}_g, \mathbf{n}_a$ are white noise terms.

The biases are usually modeled as random walks:

$$\dot{\mathbf{b}}_g(t) = \mathbf{n}_{wg}(t), \tag{12}$$
$$\dot{\mathbf{b}}_a(t) = \mathbf{n}_{wa}(t). \tag{13}$$

The accelerometer does not directly measure world acceleration; it measures specific force in the body frame.

5 Continuous-time kinematics

The motion model is

$$\dot{\mathbf{R}}(t) = \mathbf{R}(t)\,[\boldsymbol{\omega}(t)]_\times\,, \tag{14}$$
$$\dot{\mathbf{v}}(t) = \mathbf{g} + \mathbf{R}(t)\mathbf{f}(t), \tag{15}$$
$$\dot{\mathbf{p}}(t) = \mathbf{v}(t). \tag{16}$$

Here $[\boldsymbol{\omega}(t)]_\times$ denotes the skew-symmetric matrix associated with the angular velocity vector $\boldsymbol{\omega}(t)$, i.e., the matrix representation of the cross product with $\boldsymbol{\omega}(t)$.
Equation (14) describes orientation dynamics, (15) gives the velocity derivative in world frame, and (16) is standard position kinematics.

6 Role of the IMU

It is crucial to distinguish two different uses of the IMU data.

6.1 Role 1: propagating the newest state

When a new keyframe j arrives, raw IMU samples between t_i and t_j are used to propagate an initial guess for the new state from $\mathbf{x}_i$ to $\mathbf{x}_j$.

6.2 Role 2: building IMU factors in the optimization window

For every consecutive pair of states (i, j) currently in the active optimization window, the raw IMU samples between t_i and t_j are preintegrated into one IMU factor (and stored). Therefore preintegration is not only used for the newest interval: it exists for all active IMU intervals in the window. The optimization uses stored IMU factors on all intervals in the current window, not only on the newest state.

7 Preintegration from raw IMU samples

Consider one interval (i, j), and let $(\bar{\mathbf{b}}_{g,i}, \bar{\mathbf{b}}_{a,i})$ be the bias linearization point. For each IMU sample k in that interval, define the bias-corrected quantities

$$\tilde{\boldsymbol{\omega}}_k = \boldsymbol{\omega}_{m,k} - \bar{\mathbf{b}}_{g,i}, \qquad \tilde{\mathbf{f}}_k = \mathbf{a}_{m,k} - \bar{\mathbf{b}}_{a,i}. \tag{17}$$

Using a simple discrete propagation, the preintegrated quantities evolve as

$$\widehat{\Delta\mathbf{R}}_{k+1} = \widehat{\Delta\mathbf{R}}_k \operatorname{Exp}\big(\tilde{\boldsymbol{\omega}}_k \, \Delta t_k\big), \tag{18}$$

$$\widehat{\Delta\mathbf{v}}_{k+1} = \widehat{\Delta\mathbf{v}}_k + \widehat{\Delta\mathbf{R}}_k \, \tilde{\mathbf{f}}_k \, \Delta t_k, \tag{19}$$

$$\widehat{\Delta\mathbf{p}}_{k+1} = \widehat{\Delta\mathbf{p}}_k + \widehat{\Delta\mathbf{v}}_k \, \Delta t_k + \tfrac{1}{2}\widehat{\Delta\mathbf{R}}_k \, \tilde{\mathbf{f}}_k \, \Delta t_k^2. \tag{20}$$

with initialization

$$\widehat{\Delta\mathbf{R}}_{ii} = \mathbf{I}, \qquad \widehat{\Delta\mathbf{v}}_{ii} = \mathbf{0}, \qquad \widehat{\Delta\mathbf{p}}_{ii} = \mathbf{0}. \tag{21}$$

These equations use the raw IMU data and produce the measurement-side quantities $\widehat{\Delta\mathbf{R}}_{ij}, \widehat{\Delta\mathbf{v}}_{ij}, \widehat{\Delta\mathbf{p}}_{ij}$ associated with the interval (i, j).

8 Relative motion implied by the states

Now consider two states already present in the optimization variables: $\mathbf{x}_i$ and $\mathbf{x}_j$. From these states alone, we can compute the relative motion they imply:

$$\Delta\mathbf{R}_{ij}^{\text{state}} = \mathbf{R}_i^\top \mathbf{R}_j, \tag{22}$$

$$\Delta\mathbf{v}_{ij}^{\text{state}} = \mathbf{R}_i^\top \left(\mathbf{v}_j - \mathbf{v}_i - \mathbf{g}\,\Delta t_{ij}\right), \tag{23}$$

$$\Delta\mathbf{p}_{ij}^{\text{state}} = \mathbf{R}_i^\top \left(\mathbf{p}_j - \mathbf{p}_i - \mathbf{v}_i\,\Delta t_{ij} - \tfrac{1}{2}\mathbf{g}\,\Delta t_{ij}^2\right). \tag{24}$$

These expressions do *not* use raw IMU samples. They are computed only from the current guessed states in the optimizer. They describe the motion encoded by the pair $(\mathbf{x}_i, \mathbf{x}_j)$.

9 IMU residual

The IMU factor compares:

- the motion preintegrated from raw IMU samples, and
- the motion implied by the current state estimates.

The residual components are

$$\mathbf{r}_{\Delta R,ij} = \operatorname{Log}\left(\widehat{\Delta\mathbf{R}}_{ij}^\top \, \Delta\mathbf{R}_{ij}^{\text{state}}\right), \tag{25}$$

$$\mathbf{r}_{\Delta v,ij} = \Delta\mathbf{v}_{ij}^{\text{state}} - \widehat{\Delta\mathbf{v}}_{ij}, \tag{26}$$

$$\mathbf{r}_{\Delta p,ij} = \Delta\mathbf{p}_{ij}^{\text{state}} - \widehat{\Delta\mathbf{p}}_{ij}, \tag{27}$$

$$\mathbf{r}_{b_g,ij} = \mathbf{b}_{g,j} - \mathbf{b}_{g,i}, \tag{28}$$

$$\mathbf{r}_{b_a,ij} = \mathbf{b}_{a,j} - \mathbf{b}_{a,i}. \tag{29}$$

Stacking them gives

$$\mathbf{r}_{ij}^{\text{imu}} = \begin{bmatrix} \mathbf{r}_{\Delta R,ij} \\ \mathbf{r}_{\Delta v,ij} \\ \mathbf{r}_{\Delta p,ij} \\ \mathbf{r}_{b_g,ij} \\ \mathbf{r}_{b_a,ij} \end{bmatrix}. \tag{30}$$

The IMU residual is therefore the mismatch between two descriptions of the same motion over (i, j): one from measurements, one from the current optimization states.

10 Time interpretation and sliding-window implementation

A common source of confusion is the meaning of "current," "past," and "future." In online VIO, the active variable set is usually a sliding window

$$\mathbf{X} = \{\mathbf{x}_{k-m}, \mathbf{x}_{k-m+1}, \ldots, \mathbf{x}_k\}, \tag{31}$$

where $\mathbf{x}_k$ is the latest state at the current time, and the other states are recent past states. So $\mathbf{X}$ is usually a recent piece of the trajectory, not the entire trajectory and not true future states.

The optimizer refines all states currently in the window, including past ones, because new camera and IMU information can improve them. For example, if

$$\mathbf{X} = \{\mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5\}, \tag{32}$$

then the active IMU factors are typically

$$\hat{z}_{23}^{\text{imu}}, \qquad \hat{z}_{34}^{\text{imu}}, \qquad \hat{z}_{45}^{\text{imu}}. \tag{33}$$

The newest state $\mathbf{x}_5$ is initialized by IMU propagation from $\mathbf{x}_4$, but the optimization uses all active IMU factors in the current window (see Fig. 1).

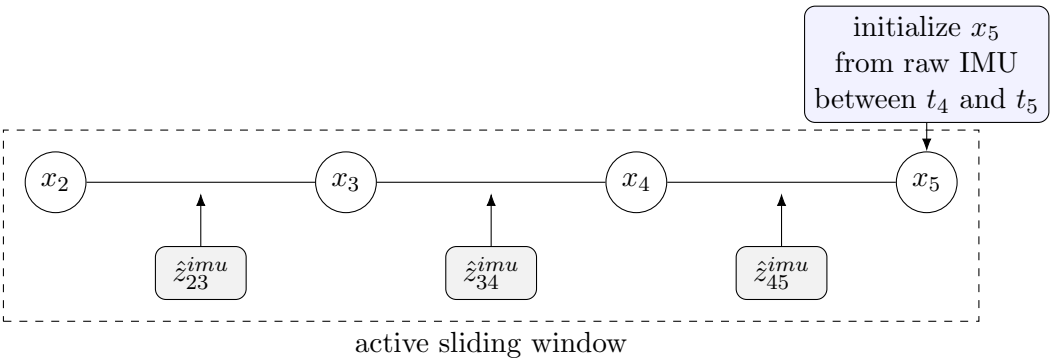

Figure 1: The active window contains recent states and all corresponding IMU factors. The newest state is initialized using raw IMU, while the optimizer uses IMU factors for every active interval.

11 Pseudo-algorithm

The newest state is initialized from IMU propagation, but the optimizer refines the full active window using all stored visual and inertial factors.

Algorithm 1 Keyframe-Based Visual-Inertial Sliding Window Optimization

At real time t_k:
Input:
$\mathcal{X}_{old} = \{x_{k-m}, \ldots, x_{k-1}\}$ ▷ Current sliding window
$\mathcal{S}_{k-1\to k} = \{(\omega_m, a_m)_t\}_{t_{k-1}}^{t_k}$ ▷ Raw IMU samples
$\mathcal{U}_k = \{\mathbf{u}_{i\ell}\}$ and $\mathcal{P} = \{\mathbf{P}_\ell\}$ ▷ 2D detections and 3D landmarks
Upon arrival of new keyframe x_k:
1. State Initialization
Propagate x_{k-1} using $\mathcal{S}_{k-1\to k}$ to obtain initial guess $\check{x}_k$
2. Window Management
$\mathcal{X} \leftarrow \{x_{k-m+1}, \ldots, x_{k-1}, x_k\}$ ▷ Slide window forward
3. Visual Residual Computation
for each observation $\mathbf{u}_{i\ell} \in \mathcal{U}_k$ **do**
  $\mathbf{r}_{i\ell}^{cam} = \mathbf{u}_{i\ell} - \pi(x_i, \mathbf{P}_\ell)$ ▷ Difference from projection model
4. Inertial Preintegration and Residuals
for each consecutive pair $(i, j) \in \mathcal{X}$ **do**
  Compute preintegrated $\Delta\hat{R}_{ij}, \Delta\hat{v}_{ij}, \Delta\hat{p}_{ij}$ from $\mathcal{S}_{i\to j}$
  Compute IMU residual $\mathbf{r}_{ij}^{imu}$ ▷ Compare state-implied motion with IMU
  Compute covariance Σ_{ij} and bias Jacobians $\mathcal{J}_{ij}$
5. Joint Optimization
$J(\mathcal{X}) = \sum \|\mathbf{r}_{i\ell}^{cam}\|_{\Sigma_{cam}}^2 + \sum \|\mathbf{r}_{ij}^{imu}\|_{\Sigma_{ij}}^2$
$\hat{\mathcal{X}} \leftarrow \arg\min_{\mathcal{X}} J(\mathcal{X})$ ▷ Solve via least-squares
6. Output
return Latest optimized state $\hat{x}_k$

12 Compact block scheme

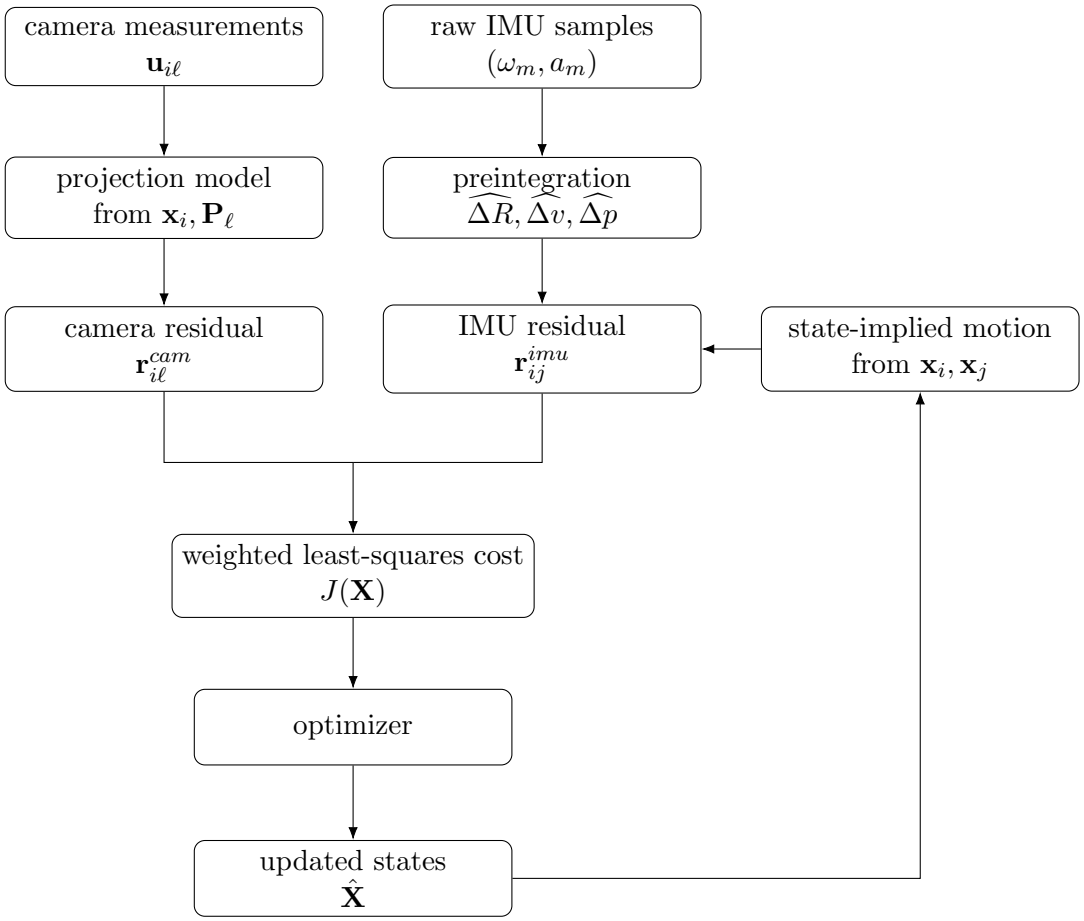

Figure 2: Visual and inertial residuals both contribute to the same optimization problem. The IMU branch explicitly compares preintegrated motion from raw measurements with motion implied by the current states.